

$$R_{\text{CoM}} = (4+5+6+7+8+4+5+6+7+8+4+5+6+7+8+4+5+6+7+8+4+5+6+7+8)/35 = 210/35 = 6$$

$$C_{\text{CoM}} = (5+6+7+8+9+10+11+5+6+7+8+9+10+11+5+6+7+8+9+10+11+5+6+7+8+9+10+11)/35 = 280/35 = 8$$

$$+8+9+10+11+5+6+7+8+9+10+11+5+6+7+8+9+10+11)/35 = 280/35 = 8$$

Here, $N=35$ (total number of pixels on the pattern's body)

Therefore $(R_{\text{CoM}}, C_{\text{CoM}}) = (6, 8)$

The centre of mass for the test

pattern (Fig. 3) comes out to be at location (6, 8), marked as white in Fig. 4.

Using the centre of mass, the pattern's shape and size can be estimated by computing the radii profile.

Some examples of centre-of-mass (marked with '+') extraction are shown in Fig. 5.

Code listing for centre-of-mass extraction using MATLAB is given alongside:

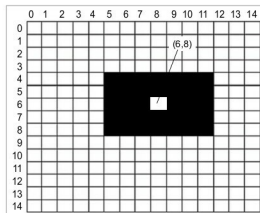


Fig. 4: Binary pattern with centre of mass at (6, 8)

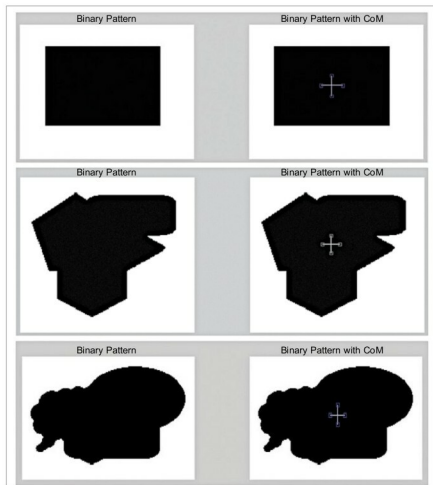


Fig. 5: Snapshots of centre-of-mass extraction

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

```
close all,
clear all,
clc,

ProjectPath = pwd;
PatternPath = strcat(ProjectPath, '\
    Pattern_3.jpg');

Orig = imread(PatternPath);
Gray = rgb2gray(Orig);
Threshold = graythresh(Gray);
BinaryPattern = im2bw(Gray, Threshold);
subplot(1,2,1);
imshow(BinaryPattern);
title('Binary Pattern');

[row col] = size(BinaryPattern);

Row_CoM=0;
Col_CoM=0;
Count=0;
for r=1:row
    for c=1:col
        if (~BinaryPattern(r,c)),
            Row_CoM = Row_CoM + r;
            Col_CoM = Col_CoM + c;
            Count = Count+1;
        end
    end
end

Row_CoM = Row_CoM/Count;
Col_CoM = Col_CoM/Count;

subplot(1,2,2);
imshow(BinaryPattern);
title('Binary Pattern with CoM');

j = imline(gca,[Col_CoM Col_CoM],
    [Row_CoM-10 Row_CoM+10]);

j = imline(gca,[Col_CoM-10 Col_CoM+10],
    [Row_CoM Row_CoM]); EFY
```



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